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A probabilistic framework based on the gradient descent algorithm for multi-objective land use optimization

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A probabilistic framework based on the gradient descent algorithm for multi-objective land use optimization

ABSTRACT

Land use planning seeks to outline the future location and type of development activity. The planning process should reconcile development with environmental conservation and other concerns pertaining to sustainability; hence multi-objective spatial optimization is considered an effective tool to serve this purpose. However, as the number of social, economic, and environmental objectives increases, especially when numerous spatial units exist, the curse of dimensionality becomes a serious problem, making previous methods unsuitable. In this paper, we formulate a probabilistic framework based on the gradient descent algorithm (GDA) to search for Pareto optimal solutions more effectively and efficiently. Under this framework, land use as decision parameter(s) in each cell is represented as a probability vector instead of an integer value. Thus, the objectives can be designed as differentiable functions such that the GDA can be used for multi-objective optimization. An initial experiment is conducted using simulation data to compare the GDA with the genetic algorithm, with the results showing that the GDA outperforms the genetic algorithm, especially in large-scale scenarios. Furthermore, the framework is shown to be effective and efficient in a real-world case study of Shenzhen. These outcomes demonstrate that the proposed GDA model is capable of generating optimal land use scenarios in a more efficient manner, rendering it a pragmatic tool for planning practices.

KEYWORDS

Spatial optimization; gradient-based optimizer; planning support system; land use planning; multi-objective optimization

1. Introduction

The past few decades have witnessed rapid urbanization globally, and the urban population worldwide will increase to approximately 2.5 billion between 2018 and 2050 (United Nations 2018). To accommodate such a swiftly growing population, urban areas in the world are expanding by approximately 108 km² every day (Seto and Pandey 2019). The rapid urban expansion has brought about numerous problems, such as environmental pollution, resource scarcity, and inadequate planning (Elmqvist *et al.* 2013). Hence, sustainability has become a goal of urban development, which seeks to strike a balance between economic development and ecosystem protection simultaneously (Stokes and Seto 2018).

Urban land use is central to the sustainability of a city because it decides the configuration of the urban area and has long-term implications for the built environment and the lives of residents (Seto and Pandey 2019). Technically, land use planning aims to rationally allocate land resource use so as to achieve the ideal of sustainability. Previous studies have modeled the task of forming an appropriate land use plan automatically and quantitatively as the land use optimization problem, which can be considered as a decision-making process that allocates different land use categories to specific areas in a region (Cao *et al.* 2011, Tong and Murray 2012, Cao and Huang 2019). In general, the process is a trade-off considering multiple objectives that tend to be in conflict with each other. For example, from the perspective of sustainability, urban

planners should consider the economic effect as well as the ecological effect, but there are conflicts between them. Compared with the ordinary optimization problem, spatial land use optimization emphasizes not only the category selection and amount but also the location (Cao *et al.* 2012). In other words, spatial land use optimization usually includes some spatial objectives which tend to be nonlinear and thus increase the complexity of the problem. Such nonlinear multi-objective optimization is in fact a nondeterministic polynomial hard (NP-Hard) problem, and its complexity depends on the number of objectives and spatial units. Given k objectives and n spatial units that can be allocated as one of m types of land use, the time complexity of solving the problem adopting a brute-force search is $O(km^n)$. The problem obviously becomes increasingly complex as the numbers of objectives, land use types, and spatial units increase. Therefore, appropriate methods should be developed to solve the complex problem, especially if it has massive spatial units.

Numerous studies have focused on land use optimization in the last few decades (e.g., Chandramouli *et al.* 2009; Huang and Zhang, 2014). In early studies, the linear programming (LP) model for land use planning was designed. The LP model provides exact solutions for linear and quadratic problems but is unsuitable for dealing with nonlinear and spatial objectives (Cao and Huang 2019). Additionally, for multi-objective optimization with a large solution space, the use of the LP model is time consuming (Xiao and Murray 2019). Hence, heuristic approaches, which provide a high-quality but not optimal solution within an acceptable time, have been introduced to solve land use optimization problems; such approaches include the simulated annealing (SA) algorithm (Aerts and Heuvelink 2002, Li and Ma 2018), the genetic algorithm (GA) (Feng and Lin 1999, Liu *et al.* 2015), ant-colony optimization (ACO) (Liu *et al.* 2012a), and particle swarm optimization (PSO) (Liu *et al.* 2012b). GAs, as a kind of biologically inspired algorithm, are the most popular heuristic methods for land use optimization (Li and Ma 2018, Mitchell 1998). The classical GA has been widely used for land use optimization (Cao *et al.* 2011, Chandramouli *et al.* 2009, Feng and Lin 1999), with one review reporting that the GA is the most frequently used heuristic, being included in more than half of studies on land use optimization (Rahman and Szabó 2021). As the algorithm develops, hybrid algorithm models have been developed by coupling the GA and other algorithms, such as those of ACO, PSO, and SA, so as to get rid of local optima and accelerate convergence (Ding *et al.* 2021). Previous studies have noted that the convergence time is unsatisfactory when there are many variables, and spatially explicit GAs have thus been designed to reduce the computational time and the fragmented effect in the generated land use map (Cao *et al.* 2012, Holzkämper and Seppelt 2007). Additionally, studies have revised the crossover and mutation operators used in the GA for land use planning to improve efficiency (Schwaab *et al.* 2018), and the paralleled GA has been applied to reduce the time required for optimization (Porta *et al.* 2013).

The basic idea in solving the multi-objective optimization problem is to seek Pareto optimal solutions (Huang *et al.* 2008). A Pareto optimal solution is a feasible solution that is not dominated by another feasible solution. If solution A dominates solution B, all the objectives of solution A are superior or equal to those of B. The Pareto front is a solution set including all of the Pareto optimal solutions. There are two basic schemes for seeking Pareto optimal solutions (Arora 2015, Huang *et al.* 2008, Li and Ma 2018). The first scheme uses methods based on Pareto dominance, which have been applied in land use optimization (Balling *et al.* 1999, Cao *et al.* 2011). These methods work simultaneously on numerous points so that the Pareto front is located directly. As an example, the Nondominated Sorting Genetic Algorithm-II (NSGA-II) proposed by Deb *et al.* (2002) is a GA that applies a Pareto-based ranking scheme to search for the nondominated set. The user can make the final decision by choosing one from the searched Pareto

front (Arora 2015, Li and Ma 2018). The second scheme, which transforms the multi-objective problem into a single-objective problem, is more intuitive. The simplest method is the weighted-sum method, in which the objectives are combined into a single objective according to weights provided by the user. Although nonconvex optimal solutions cannot be obtained using the weighted-sum method, it is still the method most commonly used in land use optimization (Aerts *et al.* 2003, Cao *et al.* 2020, Liu *et al.* 2015). Additionally, goal programming is widely adopted in land use optimization to minimize the deviation from the given goal for each objective (Li and Ma 2018, Stewart *et al.* 2004). These approaches only detect one solution on the frontier, so the user either makes the final decision after generating a set of Pareto solutions or gives prior sufficient preference information before the optimization process (Arora 2015).

Previous studies have shown the improved efficiency and effectiveness of applying heuristic algorithms to land use optimization, but there remain several limitations to the existing methods. Owing to the discrete nature of land use, the methods most commonly used in previous studies are random derivative-free optimization methods (e.g., GAs), which tend to have a possible rate of convergence that is far below that of usual schemes (Nesterov and Spokoiny 2017). In the situation that it is feasible to compute the gradient, Nesterov and Spokoiny (2017) confirmed that the iteration cost of random methods is the same as that of the gradient methods, which means that the total computational time of the random schemes is $O(n)$ times that of the gradient methods. Studies have tended to improve the efficiency of GAs by improving operators or integrating hybrid models, but the time complexity has remained limited by the nature of the random methods. In spite of the fact that random methods are the most popular methods for land use optimization, their runtimes are too long when the number of spatial units (i.e., the size of decision variables to be optimized) is large (Li and Ma 2018). As the dimensionality of the decision variables increases, the search space becomes larger and more complex. Thus, random methods require more time and computing resources to conduct the evolutionary operations. Furthermore, a larger population of candidate solutions is needed for exploring the search space, and the bundle of computation consequently increases. For methods based on comparison evaluation strategies, the convergence rate is inversely proportional to the dimensionality of the search space (Auger and Hansen 2016). Considering their high computational costs, traditional evolutionary algorithms are not suitable for solving large-scale optimization problems (i.e., optimization problems with a large number of decision parameters—usually more than 2,000). Although frontier studies have proposed methods of dealing with large-scale problems (Yi *et al.* 2020), they still do not meet the requirements of practical applications with an excessively large number of parameters, such as deep neural network optimization and land use optimization problems. For such problems with numerous decision parameters, random methods might be able to search for a satisfactory solution in theory, but the search time tends to be unacceptable. As a result of urban expansion, the number of land units to be planned will gradually increase in the future. In the context of big data for smart cities, the resolution of planned units will become finer, and the considered objectives will be more sophisticated. The existing land use optimization frameworks have limited ability to deal with such a complex problem, and it is thus necessary to build a more efficient framework.

Automatic differentiation is a technique that addresses the above problem by accurately calculating high-dimension derivatives, and it is used to improve the efficiency of evaluating the partial derivatives of functions with numerous inputs (Baydin *et al.* 2017, Nesterov and Spokoiny 2017). Thus, in the field of machine learning, gradient-based methods are much more favored when gradient computation is feasible. As part of the recent boom of deep learning, the gradient descent algorithm (GDA) has played an important role in the optimization of deep learning model

parameters. Massive modified GDA-based optimizers have been shown to be feasible and efficient in searching for suitable parameters of deep neural networks (Kingma and Ba 2014).

Given the need for an effective and efficient model that can be fulfilled by gradient-based methods, the main aim of this paper is to formulate a framework to transform the discrete multi-objective problem into a probabilistic one and apply the GDA for optimization. In contrast with previous studies, land use in a cell is not a discrete category but a vector that represents the probabilities of the cell being allocated to various categories. Hence, the spatial objective functions are designed as continuous functions, and the GDA is then adopted to solve the problem more efficiently. With the proposed framework, it is possible to seek optimal solutions for land use allocation problems with massive spatial units and complex objectives.

The remainder of this paper is organized as follows. In Section 2, we describe two frameworks and define the objectives and constraints for land use optimization. Section 3 introduces the defined loss function and the adopted GDA optimizer. A case study and simulation data experiment are reported to examine the efficiency and effectiveness of the framework in Sections 4 and 5. In Section 6, we discuss the parameter sensitivity and elucidate the strengths and limitations of the probabilistic framework. Finally, we conclude this paper in Section 7.

2. Model formulation: multi-objective optimization for land use

In building a spatial multi-objective model, the spatial data type (i.e., vector data or raster data) is important (Cao *et al.* 2020, Li and Ma 2018). In this paper, the multi-objective optimization for the land use model will be defined based on raster data, because the shapes of plots of vector data are fixed and difficult to divide (Li and Ma 2018).

2.1. Model frameworks

Assume that a land use raster image has $M \times N$ pixels, where each pixel is assigned one of K land use types. The image is thus considered as a three-dimensional array $\mathbf{P}_{M \times N \times K}$, and $p_{i,j,k}$ denotes the element located in the i^{th} row and j^{th} column of the k^{th} band in the image $\mathbf{P}$. The aim of land use optimization is to find an optimal three-dimensional array $\mathbf{P}^*$ that maximizes the benefits or minimizes the costs according to given objective functions.

2.1.1. Discrete framework

In previous studies, the discrete multi-objective optimization for land use (MOLU) model (Cao *et al.* 2011) has been the framework most commonly used to formulate the optimization problem. The discrete MOLU model is thus formulated as follows:

$$\begin{aligned} \min_{\mathbf{P}} \quad & f^{(o)}(\mathbf{P}), \forall o \in O \\ \text{s. t.} \quad & \sum_{k=1}^K p_{i,j,k} = 1, \forall i, j \\ & p_{i,j,k} \in \{0,1\}. \end{aligned}$$

In this discrete framework, O is a set of objectives to be optimized, $f^{(o)}$ is the objective function for a specific objective o in O , and $p_{i,j,k}$ is a binary variable such that $p_{i,j,k} = 1$ when the pixel (i, j) is allocated the k^{th} land use category and $p_{i,j,k} = 0$ otherwise.

The discrete MOLU model is straightforward, but the parameters are discrete. Thus, the objective functions are not continuous, which means that it is impossible to calculate the derivatives. Due to this characteristic of objective functions, the discrete MOLU model is not suitable for gradient-based methods. Consequently, gradient-free methods are the methods most adopted to optimize the discrete MOLU model.

2.1.2. Probabilistic framework

To apply the gradient-based optimizer in optimizing spatial land use, a probabilistic multi-objective optimization of land use (pMOLU) model is proposed:

$$\begin{aligned} \min_{\mathbf{P}} \quad & f^{(o)}(\mathbf{P}), \forall o \in O \\ \text{s. t.} \quad & \sum_{k=1}^K p_{i,j,k} = 1, \forall i, j \\ & 0 \leq p_{i,j,k} \leq 1. \end{aligned}$$

In the proposed probabilistic framework, $p_{i,j,k}$ is not a discrete variable as before but a continuous variable ranging from 0 to 1, such that the loss function is differentiable. From the perspective of probability theory, $p_{i,j,k}$ is considered as the probability that the pixel (i, j) is allocated the k^{th} land use category.

In the discrete framework mentioned in Section 2.1.1, each cell in the raster is allocated one unique land use category. However, in the proposed probabilistic modeling, the parameters of each cell are defined as a probability vector so that a differentiable objective function can be defined. Thereby, the continuous decision variables can be determined using gradient-based optimization methods. Based on the optimal probability vector, the land use category with maximum probability is allocated to the corresponding cell.

2.2. Objectives

The objectives for the land use planning model are defined in the form of mathematical equations. We designed seven differentiable objective functions that can be applied to the pMOLU model to generate land use plans, and they are adopted in the case study presented in Section 5. It is noted that any differentiable functions can be used as the objective functions for pMOLU.

2.2.1. Maximizing the economic benefit

The economic benefit is one of the most important objectives in urban land use planning. Urban planners tend to modify the land use structure and distribution of a city to maximize the utilization of the land. In this study, we design a function summarizing the contributions made by all the pixels to urban economic benefit with consideration of the marginal utility and spatial suitability.

The simplest but most used definition of the objective function for economic benefit is the calculation of the weighted sum of the quantity of each land use type.

Let $\mathbf{q}^{(\mathbf{P})} = [q_1, q_2, \dots, q_K]^T$ denote a vector representing the land use structure for a raster $\mathbf{P}_{M \times N \times K}$. In $\mathbf{q}^{(\mathbf{P})}$, q_k is the proportion of pixels allocated the k^{th} land use category in $\mathbf{P}$:

$$q_k = \frac{1}{MN} \sum_{j=1}^N \sum_{i=1}^M p_{i,j,k},$$

where $p_{i,j,k}$ is an element in $\mathbf{P}$. Obviously, $0 \leq q_k \leq 1$.

The planner predefines a K -dimensional weight vector $\mathbf{w} = [w_1, w_2, \dots, w_K]^T$, where w_k is the weight representing the benefit per unit of the k^{th} land use category. The weighted-sum objective function definition is then written as follows:

$$f^{(o)}(\mathbf{P}) = \mathbf{w}^T \mathbf{q}^{(\mathbf{P})}. \quad (1)$$

However, the above definition is linear and does not reflect reality. In fact, the relation between the economic benefit and the quantity of allocated land use type is not linear like in Equation (1). According to the law of diminishing marginal utility (DMU) (Mankiw 2014), given fixed levels of the population and technology, when the percentage of one specific land use increases, the marginal economic benefit yielded by the additional land unit should decline. As an example, the economic benefit of additional commercial land when there is little commercial land should be higher than that when there is excessive commercial land. From the perspective of model design, the introduction of DMU is considered as a regularization of the objective.

Supposing that the proportion of a specific land use type is q and given a marginal function $u(q)$, the equivalent benefit function (or average utility) is the following:

$$\bar{w}(w, q) = \frac{w}{q} \int_0^q u(q) dq,$$

which is considered as the average benefit utility per unit when the proportion of specific land use is q .

According to the law of DMU, the marginal function $u(q)$ should monotonically decrease and there should be no diminishing effect when $q = 0$. In this paper, the adopted marginal function is $u(q) = 1 - q^2$, and we then have $\bar{w}(w, q) = w(1 - \frac{1}{3}q^2)$ (see Appendix A for the derivation).

For all land use categories, the equivalent benefit vector $\bar{\mathbf{w}}$ is calculated as follows:

$$\bar{\mathbf{w}}(\mathbf{w}, \mathbf{q}^{(\mathbf{P})}) = [\bar{w}(w_1, q_1), \bar{w}(w_2, q_2), \dots, \bar{w}(w_K, q_K)]^T.$$

Considering the DMU, Equation (1) is modified as follows:

$$f^{(o)}(\mathbf{P}) = \bar{\mathbf{w}}(\mathbf{w}, \mathbf{q}^{(\mathbf{P})})^T \mathbf{q}^{(\mathbf{P})}. \quad (2)$$

Although Equation (2) is well considered, the spatial heterogeneity is still neglected. Taking the spatial suitability into account, a spatial weight raster $\mathbf{S}_{M \times N}$ with the same size as raster $\mathbf{P}$ should be given. The value of each pixel of $\mathbf{S}$ represents the suitability of the pixel. For instance, the value could be the distance to the nearest road, for economic benefits are less when their access is inconvenient. The spatial weight of pixel (i, j) is $s_{i,j}$ (ranging from 0 to 1), and we define $\mathbf{r}^{(\mathbf{P}, \mathbf{S})} = [r_1, r_2, \dots, r_K]^T$, where r_k is a spatially weighted land use quantity whose definition is similar to q_k :

$$r_k = \frac{1}{MN} \sum_{j=1}^N \sum_{i=1}^M s_{i,j} p_{i,j,k}.$$

Finally, the objective function is defined as follows:

$$f^{(o)} = \bar{\mathbf{w}}(\mathbf{w}, \mathbf{q}^{(\mathbf{P})})^T \mathbf{r}^{(\mathbf{P}, \mathbf{S})}. \quad (3)$$

The objective function of the economic benefit is expressed as follows:

$$f^{(economic)}(\mathbf{P}) = \overline{\mathbf{w}}(\mathbf{w}^{(economic)}, \mathbf{q}(\mathbf{P}))^T \mathbf{r}(\mathbf{P}, \mathbf{S}^{(economic)}). \quad (4)$$

2.2.2. Maximizing the ecological benefit

Environmental sustainability is an indispensable objective in land use planning. In this paper, the ecological benefit is designed as one of the objectives, so as to reduce industrial pollution and maintain the ecological environment in the study area.

Similar to the case for the economic benefit function, we use the spatial weighted diminishing model formulated in Equation (3) to summarize the spatial-relative ecological benefit of all pixels to define the ecological benefit objective function:

$$f^{(ecological)}(\mathbf{P}) = \overline{\mathbf{w}}(\mathbf{w}^{(ecological)}, \mathbf{q}(\mathbf{P}))^T \mathbf{r}(\mathbf{P}, \mathbf{S}^{(ecological)}). \quad (5)$$

2.2.3. Maximizing the residential housing capacity

Most cities face the problem of a lack of residential housing. As the city develops, the population increases and housing becomes scarce, and it is thus necessary to allocate more residential areas and to use land intensively.

The law of DMU is not applicable because the housing capacity correlates only to the physical space, and the residential land proportion does not vary. A linear model is thus adopted to aggregate the housing capacity provided in every pixel:

$$f^{(housing)}(\mathbf{P}) = \mathbf{w}^{(housing)}^T \mathbf{q}(\mathbf{P}). \quad (6)$$

2.2.4. Maximizing the employment capacity

Besides the housing capacity, the employment capacity is a basic function of cities (Huang and Zhang 2014). Sufficient employment capacity is required to provide enough jobs for citizens. Similar to the case for the housing capacity, the objective function is the total of employment capacity provided by all the pixels:

$$f^{(employment)}(\mathbf{P}) = \mathbf{w}^{(employment)}^T \mathbf{q}(\mathbf{P}). \quad (7)$$

2.2.5. Maximizing livability

City livability, as a vital urban issue, is pursued in urban planning to provide citizens a livable environment (Kaal 2011). Livability reflects the quality of life of citizens in terms of the built environment, natural environment, and living convenience.

In this paper, living convenience and ecological accessibility are considered as measures of livability. In a region, living convenience can be defined as the number of shops (relative to the area of commercial land) divided by the number of residents, whereas ecological accessibility can be defined as the ecological benefits divided by the number of residents. The adopted objective represents the how convenient the residents can access living services and the natural environment.

Considering spatial heterogeneity, the ecological benefits in Equation (5) can be considered as the sum of ecological benefits contributed by each cell: $f^{(ecological)}(\mathbf{P}) =$

$\sum_{v \in V^{(ecological)}} v$, where $V_{M \times N}^{(ecological)}$ is a raster, whose value at the location (i, j) is denoted as $v_{i,j}^{(ecological)}$ representing the ecological benefits contributed by the respective cell:

$$v_{i,j}^{(ecological)} = \sum_{k=1}^K \bar{w}(w_k^{(ecological)}, q_k) s_{i,j}^{(ecological)} p_{i,j,k}.$$

Similarly, the cell values in $V_{M \times N}^{(convenient)}$ and $V_{M \times N}^{(housing)}$ are obtained as follows:

$$v_{i,j}^{(convenient)} = \sum_{k=1}^K w_k^{(convenient)} s_{i,j}^{(convenient)} p_{i,j,k},$$

$$v_{i,j}^{(housing)} = \sum_{k=1}^K w_k^{(housing)} p_{i,j,k}.$$

Mean filtering is then applied to calculate the mean amounts of the nearby commercial area, ecological benefits, and the residential housing capacity. Let $h_M^{(r)}$ denote an $r \times r$ mean filter whose elements are all $\frac{1}{r^2}$. The raster V can be mapped to a new raster $\tilde{V}$:

$$\tilde{V} = V * h_M^{(r)}, \quad (8)$$

where $*$ denotes the convolution operator.

The objective function of livability is thus defined as follows:

$$f^{(livable)}(\mathbf{P}) = \sum_{j=1}^N \sum_{i=1}^M \frac{\tilde{v}_{i,j}^{(ecologic)} + \tilde{v}_{i,j}^{(convenient)}}{\tilde{v}_{i,j}^{(housing)} + 1}, \quad (9)$$

where $\tilde{v}_{i,j}^{(o)}$ is the (i, j) element in $\tilde{V}^{(o)}$ for measurement o .

2.2.6. Minimizing the employment–housing divergence

The employment–housing (E-H) balance, is an important index for estimating commuting efficiency (Cervero 1989). The spatial distribution of the residential area and workplaces determines the E-H balance, which can be achieved by allocating a residential area near the land use that provides jobs (Zhou *et al.* 2017).

In this paper, to maintain the E-H balance, the capacity divergence between employment and housing is supposed to be minimized. That is, the objective function is defined as follows:

$$f^{(EH)}(\mathbf{P}) = \sum_{j=1}^N \sum_{i=1}^M |\tilde{v}_{i,j}^{(housing)} - \tilde{v}_{i,j}^{(employment)}|. \quad (10)$$

The objective function represents the difference between the number of residents and the employment positions provided within a specific distance to the residents. A lower objective value indicates a better balance between employment and housing.

2.2.7. Maximizing compatibility

In practice, a given land use is usually distributed close to zones with a nonconflicting land use type (Huang and Zhang 2014). As an example, an ecological zone is seldomly located near a zone of heavy industry. Furthermore, areas of the same land use tend to cluster together, as is desired in land use planning (Cao and Huang 2010). In other words, land use has preferences

in terms of neighborhoods. In this study, we use a compatibility index to measure these preferences. High compatibility means that the land use of a pixel is less conflicting with that of nearby pixels. To calculate the index value, the planner should provide a compatible matrix $\mathbf{C}_{K \times K}$, whose element $c_{i,j}$ is the degree of compatibility between the i^{th} land use and j^{th} land use. A higher degree of compatibility means that it is more likely that the two land use categories are allocated together. It is natural that $0 \leq c_{i,j} \leq 1$, $c_{i,j} = c_{j,i}$.

Let $\mathbf{P}_{(k)}$ denote the k^{th} band of the land use planning $\mathbf{P}_{M \times N \times K}$. The matrix $\mathbf{P}_{(k)}$ with M rows and N columns is a map of the probability of allocating the k^{th} land use. Using mean filtering in Equation (8), a matrix $\tilde{\mathbf{P}}_{(k)}$ is calculated for land use k . By stacking K raster images, we have an array $\tilde{\mathbf{P}}_{M \times N \times K}$ whose k^{th} band is $\tilde{\mathbf{P}}_{(k)}$.

Hence, the compatibility μ of the pixel in the i^{th} row and j^{th} column is calculated as follows:

$$\mu_{i,j} = \sum_{k_1=1}^K \sum_{k_2=1}^K c_{k_1,k_2} \tilde{p}_{i,j,k_1} \tilde{p}_{i,j,k_2} = \tilde{\mathbf{p}}_{i,j}^T \mathbf{C} \tilde{\mathbf{p}}_{i,j},$$

where $\tilde{\mathbf{p}}_{i,j} = [\tilde{p}_{i,j,1}, \tilde{p}_{i,j,2}, \dots, \tilde{p}_{i,j,K}]^T$.

Finally, the objective function of compatibility is expressed as follows:

$$f^{(compatibility)}(\mathbf{P}) = \sum_{j=1}^N \sum_{i=1}^M \mu_{i,j}. \quad (11)$$

2.3. Constraints

2.3.1. Conversion cost

The conversion cost should be considered when changing the land use. The conversion cost includes the cost of removing or reconstructing existing buildings and other structures, and the cost of building new facilities for newly allocated land use. In this study, we calculate the total cost based on the land use transformation matrix counting the differences between the optimized and original scenarios.

Supposing that the original land use raster (for the current situation) is $\mathbf{P}'$, the conversion cost is a measure of the difference between a solution raster $\mathbf{P}$ and the original raster $\mathbf{P}'$.

Here, we define a flattening function $\mathcal{F}$ to reshape the land use raster $\mathbf{P}_{M \times N \times K}$ to a $MN \times K$ matrix $\mathcal{F}(\mathbf{P})$, where the vector in the $(i \cdot j)^{\text{th}}$ column is the land use probability vector $\mathbf{p}_{i,j} = [p_{i,j,1}, p_{i,j,2}, \dots, p_{i,j,K}]^T$ for pixel (i,j) .

The transformation matrix $\mathbf{T}$ is calculated as follows:

$$\mathbf{T} = \mathcal{F}(\mathbf{P}')^T \mathcal{F}(\mathbf{P}).$$

The element $t_{i,j}$ in the i^{th} row and j^{th} column of $\mathbf{T}$ denotes the number of pixels converted to the j^{th} land use from the i^{th} land use.

Considering that the conversion cost depends on the allocated land use type and original land use type, a conversion cost weight matrix $\mathbf{\Delta}$ should be predefined. The element $\delta_{i,j}$ of the i^{th} row and j^{th} column of matrix $\mathbf{\Delta}$ represents the cost of a cell changing from the i^{th} land use to the j^{th} land use.

The function with which to measure the conversion cost is defined as follows:

$$f^{(conversion)}(\mathbf{P}) = \sum_{j=1}^K \sum_{i=1}^K \delta_{i,j} t_{i,j}. \quad (12)$$

The conversion cost can be set as a constraint by defining a threshold bound as the allowable range for the feasible set. To make the conversion cost comprehensible, we normalize the conversion cost function by its maximum value $f^{(conversion)*}$:

$$f^{(CN)}(\mathbf{P}) = \frac{f^{(conversion)}(\mathbf{P})}{f^{(conversion)*}}.$$

Thus, a threshold $\tau^{(CN)}$ can be predefined in the range from 0 to 1. When the normalized conversion cost $f^{(CN)}$ is larger than the threshold $\tau^{(CN)}$, the optimizer stops the iteration.

2.3.2. Restricted area

There are spaces reserved for special purposes, such as military purposes, ecological protection, and the protection of cultural relics. A reserved space is not allowed to be developed, so a raster map of such restricted areas should be provided before optimization.

Therefore, before evaluating the objective value of a specific planning solution $\hat{\mathbf{P}}$, the probability vector of a restricted cell should be set as $\mathbf{0}$. Supposing that $\mathcal{R}$ denotes a raster map of these restricted areas, a pixel in $\mathcal{R}$ takes a value of 0 when the pixel is restricted for development and a value of 1 otherwise. The value of objective o for $\hat{\mathbf{P}}$ is then $f^{(o)}(\mathcal{R} \circ \hat{\mathbf{P}})$, where $\circ$ is the pixel-wise multiplication operator. Finally, the optimal solution is expressed as follows:

$$\hat{\mathbf{P}}^* = \mathcal{R} \circ \arg \min_{\mathbf{P}} f^{(o)}(\mathcal{R} \circ \mathbf{P}). \quad (13)$$

3. Gradient-based optimization

Within the probabilistic framework, we formulate the objectives as differentiable functions such that the GDA can be applied to find optimal solutions. In this section, we define the loss function using the utility function method and also introduce the adopted GDA optimizer named Adaptive Moment Estimation (Adam).

3.1. Loss function

In applying the GDA, a loss function should be designed first. Thus, we define the loss function for GDA based on the utility function method, which is a preference-based method for solving the multi-objective problem. The method maps the objectives of a feasible solution on a scalar value with a multi-objective utility function (MOUF) so as to convert the objectives into a single objective (Huang *et al.* 2008). The planner can predefine the parameters of the MOUF to combine different objectives, or systematically tune the parameters to seek various nondominated solutions one by one (Huang *et al.* 2008, Li and Ma 2018).

Supposing that all of the objective functions are additive and positive, the most common MOUF is the weighted-sum function that uses a preference vector for weighting each objective. A more general definition of the MOUF is based on the distance between a solution $\mathbf{P}$ and utopia point $\mathbf{P}^*$. The utopia point $\mathbf{P}^*$ might not exist, but its objective values are all we need to consider. Thus, we can simply define its objective values such that $f^{(o)}(\mathbf{P}^*) \geq f^{(o)}(\mathbf{P}), \forall o \in O, \forall \mathbf{P}$. Without loss of generality, $f^{(o)}(\mathbf{P}^*)$ can be set as the optimal objective value when o is the sole objective. The general MOUF is defined by the Minkowski distance in L_p space ($p \in \mathbb{Z}, p \geq 1$):

$$f_{p,\lambda}^{(MOU)}(\mathbf{P}) = \left[\sum_{o \in O} \lambda^{(o)} [f^{(o)}(\mathbf{P}) - f^{(o)}(\mathbf{P}^*)] \right]^{\frac{1}{p}}, (\lambda^{(o)} > 0). \quad (14)$$

Obviously, when $p = 1$, the general MOUF is actually the weighted-sum MOUF. When $2 \leq p < \infty$, the function is nonlinear and not easy to handle, and it is thus used rarely. When $p \rightarrow \infty$, Equation (14) is equivalent to the min-max MOUF:

$$f_{\lambda}^{(MOU)}(\mathbf{P}) = \max_{o \in O} [\lambda^{(o)} [f^{(o)}(\mathbf{P}) - f^{(o)}(\mathbf{P}^*)]], (\lambda^{(o)} > 0). \quad (15)$$

Das and Dennis (1997) proved that the minimum of the weighted-sum MOUF is the Pareto optimal solution, but their approach cannot find the solution on the nonconvex part of the Pareto front (Huang *et al.* 2008, Li and Ma 2018). Using the min-max MOUF, these weak nondominated solutions can also be captured, as proved in Appendix B. Therefore, the min-max MOUF is adopted to construct the loss function in this paper. The loss function for optimization is defined as follows:

$$\mathcal{L}_{\lambda}(\mathbf{P}) = \max_{o \in O} \left[\frac{1}{\lambda^{(o)}} \frac{|f^{(o)}(\sigma(\mathbf{P})) - f^{(o)}(\mathbf{P}^*)|}{|f^{(o)}(\mathbf{P}') - f^{(o)}(\mathbf{P}^*)|} \right], (\lambda^{(o)} > 0), \quad (16)$$

where $\mathbf{P}'$ is the original land use plan (i.e., the original land use patterns).

In Equation (16), the softmax function $\sigma: \mathbb{R}^K \rightarrow \mathbb{R}^K$ is applied:

$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, i = 1, 2, \dots, K.$$

In the gradient-based optimization, the optimized parameters change according to the gradient of the loss function. That is to say, some components of the land use vector for a pixel can be negative or greater than 1, and the land use vector might not sum to 1. Hence, the softmax function converts a K -dimensional vector into a K -dimensional probability vector, to ensure the vector of each pixel in the solution can be interpreted as a probability vector.

3.2. Optimizer

The GDA is an optimization algorithm that finds a local optimal of a differentiable function and has been widely applied in searching for the parameters of complex models, such as those of deep learning (Ruder 2016). With the rapid development of deep learning and graphics processing unit (GPU) accelerating techniques, many modified GDA-based optimizers have been shown to be efficient and feasible for the optimization of deep learning model parameters. In this paper, we thus apply the GDA to land-use optimization so that an optimal solution can be found in less time.

Suppose that the optimization problem is to find a set of parameters $\hat{\boldsymbol{\theta}}$ that minimize the loss function $\mathcal{L}(\boldsymbol{\theta})$:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}).$$

The idea of the GDA is to iteratively change the parameters according to the gradient:

$$\boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_{t-1} - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_{t-1}),$$

where θ_t denotes the parameters at the t^{th} iteration, η is the learning rate, and $\nabla\mathcal{L}(\theta)$ is the gradient of $\mathcal{L}$ at point θ . The iteration stops when θ converges to a stable result.

The normal GDA, also called the steepest-descent method, is intuitive but has problems. In the steepest-descent method, parameters are updated based on only the gradient. Therefore, the convergence is slow in the direction having a small gradient, and the parameters cannot be updated when the gradient is close to zero, which means the updating ceases at local minima or saddle points. Furthermore, in ravine areas where the gradient is much steeper in one dimension than in another dimension, the method oscillates in the steep direction (Ruder 2016).

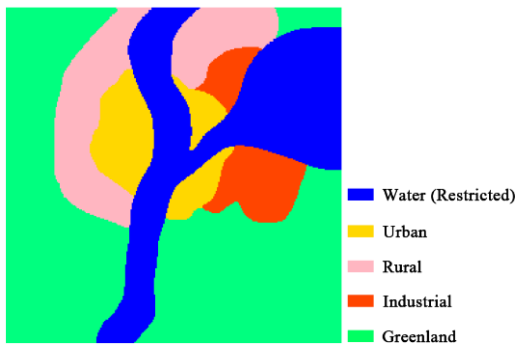
The Momentum optimizer (Qian 1999) has been proposed to solve the above problem. The Momentum optimizer updates the parameters according to not only the gradient but also the parameter increments in the last iteration. With the Momentum optimizer, the parameters are updated more rapidly if the gradient is almost the same as the last gradient, and the training speed is thus improved. If the gradient direction is opposite the last gradient direction, the parameter increments do not change suddenly, which helps get rid of the local minima and saddle points.

Some optimizers, such as AdaGrad (Duchi *et al.* 2011), applied the adaptive learning rate to help improve convergence speed and performance. The adaptive learning rate enables GDA to adapt the learning rate to various parameters such that the step length of the parameter update is larger in a steeper direction.

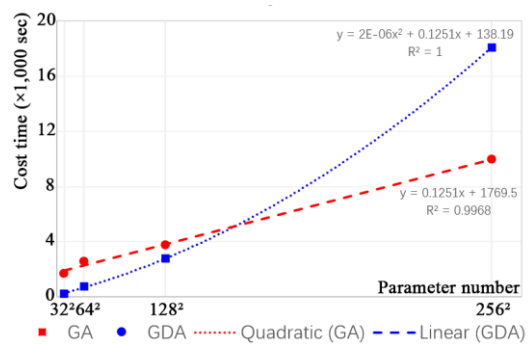
Combining both the momentum and adaptive learning rate, a widely applied GDA method named Adam was proposed by Kingma and Ba (2014). The following are its equations for parameter updating:

$$\begin{aligned} \mathbf{v}_t &\leftarrow \beta_1 \mathbf{v}_{t-1} + (1 - \beta_1) \nabla \mathcal{L}(\theta_{t-1}), & \tilde{\mathbf{v}}_t &\leftarrow \frac{\mathbf{v}_t}{1 + \beta_1^t}, \\ \mathbf{s}_t &\leftarrow \beta_2 \mathbf{s}_{t-1} + (1 - \beta_2) \nabla \mathcal{L}(\theta_{t-1}) \circ \nabla \mathcal{L}(\theta_{t-1}), & \tilde{\mathbf{s}}_t &\leftarrow \frac{\mathbf{s}_t}{1 + \beta_2^t}, \\ \theta_t &\leftarrow \theta_{t-1} - \frac{\eta \tilde{\mathbf{v}}_t}{\sqrt{\tilde{\mathbf{s}}_t + \epsilon}}. \end{aligned}$$

Here, the momentum $\mathbf{v}_t$ and exponentially decaying average of past gradients $\mathbf{s}_0$ are initialized as $\mathbf{0}$ when $t = 0$. Note that the parameters are updated by the bias-corrected first momentum $\tilde{\mathbf{v}}_t$ and second moment $\tilde{\mathbf{s}}_t$ in Adam. We adopt Adam as the optimizer for optimization.



(a) Original land use.



(b) Time for optimization.

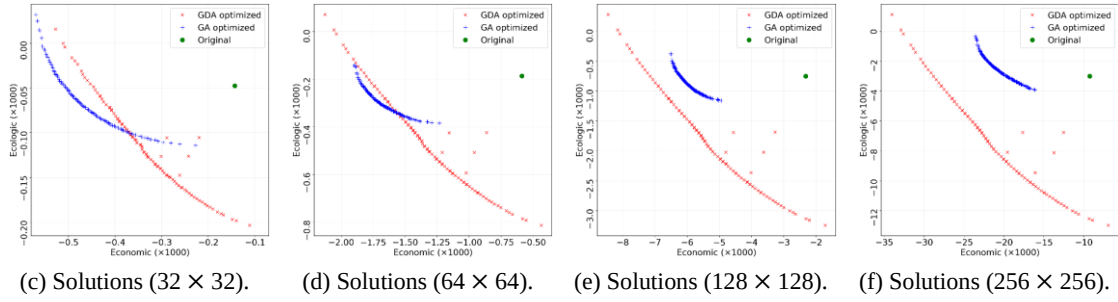


Figure 1. Simulation data and experimental results.

4. Experiment Using Simulation Data

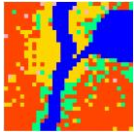
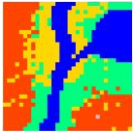
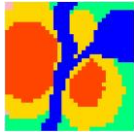
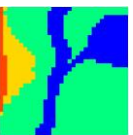
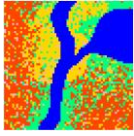
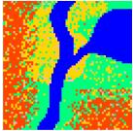
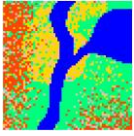
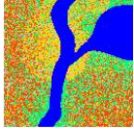
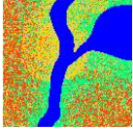
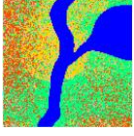
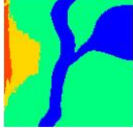
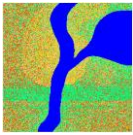
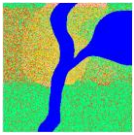
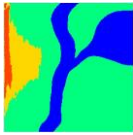
We conducted an experiment to compare the GDA and GA for the simulated land use raster with 256×256 pixels shown in **Figure 1(a)**. There were four alterable land use categories of the simulation data besides the water area, which as a restricted area was unable to change. The economic benefit and ecological benefit formulated as Equations (4) and (5) were the two objectives considered. The running environment and hyperparameters of the case study are presented in Appendix C.

The NSGA has been widely applied in land use optimization, and NSGA-III was thus adopted in this experiment for GA optimization. We implement the algorithm using a Python module named Geatpy, which is a high-performance GA toolbox (Jazzbin 2020). To evaluate the efficiencies of the GA and GDA in the situation with various sizes of the solution space, the simulated raster was resampled into three other rasters with dimensions of 32×32 , 64×64 , and 128×128 . For each raster, the GA iterated for 10,000 generations to find a nondominated population with 100 individuals, and the GDA found 100 solutions with various preference vectors by iterating the optimizer 10,000 times for each solution. These two methods both need predefinition, i.e., the reference points for NSGA-III and the utopia point for the proposed framework with GDA. They can be manually defined by the user, or generated by specific methods. In this experiment, the reference point is generated by Deb and Jain's Method (Tian *et al.* 2018), while the utopia point is defined based on the optimal values of various objectives estimated by GDA.

Figure 1(b) shows the time required by the GA and GDA to find 100 solutions. It should be noted that the consumed time includes the time for predefinition. With a small number of parameters to be optimized, the GA took less time than the GDA. However, as the number of parameters increased, the computational time of the GA increased more rapidly than that of the GDA. For the raster with 256×256 pixels, the GA took 18,091.33 seconds to obtain 100 solutions whereas the GDA took only 9,966.84 seconds. In **Figure 1(b)**, the relation between the computational time of the GDA and parameter number is linear, whereas the time cost curve of the GA is close to being quadratic. The results also concur with the view of Nesterov and Spokoiny (2017) that the total computational time of the random schemes is $O(n)$ times that of the gradient methods.

Figure 1(c) to **(f)** shows that the 100 generated solutions constitute a curve that can be seen as the Pareto front found by the algorithms. When the solution space was small, from the perspective of objective values, the Pareto front found by the GDA was no better than that found by the GA. As the solution space grew, the GA failed to find a satisfying Pareto front within 10,000 generations whereas the GDA performed well.

Table 1. Optimal scenarios and objective values for various weight vectors

Image Size	GA			GDA		
	Econ-pre	Trade-off	Eco-pre	Econ-pre	Trade-off	Eco-pre
32						
	569/32	468/70	227/114	528/16	353/111	112/204
64						
	1897/142	1681/307	1230/385	2128/72	1404/436	436/814
128						
	6511/376	5977/890	4938/1161	8456/280	5572/1739	1716/3244
256						
	23562/338	20767/2594	16124/3911	34052/1131	22516/6980	6937/12987
 Water (Restricted)  Urban  Rural  Industrial  Greenland						

^[1] The values below a solution image are the economic/ecological benefit values of the solution.

^[2] Abbreviations: Econ, Economic benefit; Eco, Ecological benefit; pre, preferred.

Table 1 gives solutions at the Pareto front obtained by the algorithms. For the image in each scalar, an economically preferred solution, ecologically preferred solution, and trade-off solution are given in the table. The table shows that the pepper noise problem was serious in the scenarios generated by the GA, and the problem was worse in the solutions for the larger images. In contrast, the GDA performed stably and well for images of all size. NSGA-III adopts the discrete operator which selects pixels randomly for mutation and crossover to explore the solution space, whereas GDA updates the variables to seek solutions based on the gradient of the spatial objectives. Therefore, the GDA can generate smoother scenarios with the spatial information given by the gradient.

5. Case study

The experiment in Section 4 demonstrated the efficiency and effectivity of the pMOLU with GDA, but a case study is necessary to evaluate the performance of proposed framework in real-world application. The Guangming district in Shenzhen was thus selected as a case study. This section describes the case study, gives the results of the study, and presents an analysis and discussion of the results.

5.1. Study area and land use classification system

The Guangming district, which is a newly established formal administrative district in Shenzhen, was selected as the study area. Guangming, as an ecological high-tech industrial district, is considered an important node of the Guangzhou–Shenzhen–Hong Kong Science and Technology Innovation corridor and a core part of the Guangdong–Hong Kong–Macao Greater Bay Area. Guangming, which has a population of 625,000, is located in the northwest of Shenzhen. Six subdistricts constitute the district with a total area of approximately 156.1 km². The area is not yet completely developed, and it is thus a suitable case study for land use optimization.

In terms of land use planning, the categorization included the 12 types of land use listed in **Table 2**. With the given land use categorization system, the land use data of Guangming district for 2014 are depicted in **Figure 2** as 850 × 804 pixels with a resolution of 20 meters × 20 meters. Excluding the pixels outside the study area and in the restricted area, there were a total of 302,422 pixels. The land use scenario in **Figure 2** was considered as the current situation and was set as the initial planning scenario for the optimization model.

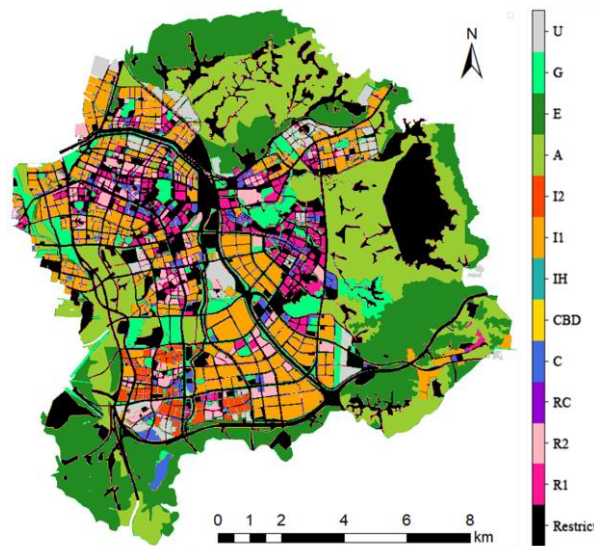


Figure 2. The Initial land use scenario.

Table 2. Land use categories in the case study

Land use	Abbreviation	Digital Number
Low-density Residential	R1	1
High-density Residential	R2	2
Residential-Commercial Mixture	RC	3
Commercial	C	4
Central Business District	CBD	5
High-tech Industrial	IH	6
Low-pollution Industrial	I1	7
High-pollution Industrial/Warehouse	I2	8
Agricultural	A	9
Ecological Zone	E	10
Green Land	G	11
Undeveloped	U	12

5.2. Results

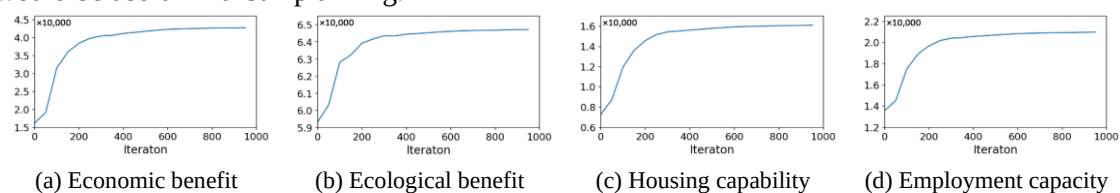
The running environment and hyperparameters of the case study are presented in Appendix C. The experiment results, including the convergence of the optimization process and the land use scenarios for the study area obtained by optimization, are described in this section.

5.2.1. Convergence

In analyzing the running time and convergence of the model, the conversion cost constraint was not considered such that the optimizer continued iterating. The optimizer iterated 10,000 times, and the preference values (i.e., λ in Equation (16)) for all objectives were set at 1.

The implemented model required 813.12 seconds for 10,000 iterations (i.e., approximately 0.08 seconds per iteration). With GPU acceleration, the model required 128.01 seconds for 10,000 iterations (i.e., 0.01 seconds per iteration). Through iteration, the objective values approached the utopia point in the optimal direction, and the values became stable at the 1,000th iteration. Compared with the original scenario, the solution had more optimal values for all objectives. Specifically, the economic benefit, ecological benefit, housing capability, employment capability, livability, and compatibility increased by 187.92%, 11.26%, 149.35%, 167.06%, 11.20%, and 2.81%, respectively, through 10,000 iterations. The E-H divergence, which was to be minimized, decreased by 9.1%. That is, the generated Pareto solution dominated the original land use. Additionally, the conversion cost, which was not an objective but a constraint indicator, was 0.1548 after 10,000 iterations.

Figure 3 presents the changes in objective values against the number of iterations. It is seen that the model converged at approximately the 400th iteration. In other words, it took approximately 32.52 seconds (5.12 seconds with a GPU) to obtain an optimal solution, which represents a time savings. After the convergence, there were no obvious changes in the objective values. It is reasonable to infer that the solution was approaching to Pareto optimal because it reached a situation in which no change improved all objectives. It should be noted that the Pareto front is based on the loss function, indicating that the influence of the selected of the utopia point on the results. In this case study, the utopia point is estimated via single-objective optimization using GDA. Since local optimal, saddle point, and the plateau of an objective might trap the optimizers, we suggest using GDA with a heuristic technique such as Adam adopted in this case. Like many other heuristic methods, it is intractable to prove the results are global optima, especially for complex objectives. Nevertheless, the convergence indicates that the proposed method successfully generates a significantly improved land use scenario efficiently, which would be useful in urban planning.



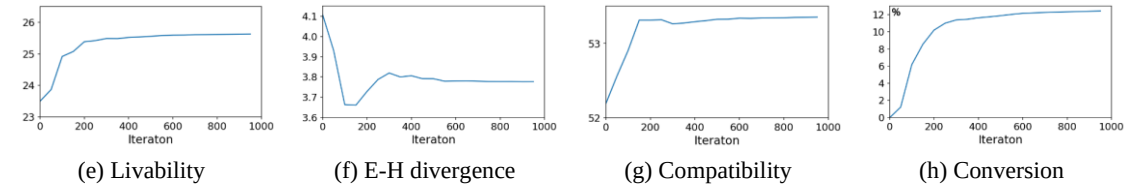


Figure 3. Convergence of objective values with iteration.

5.2.2. Optimal scenarios

To test the effects of preference values on the results, various scenarios were generated using different preference weights. The conversion cost threshold $\theta^{(CN)}$ and maximum iteration times were set at 0.1 and 1,000 correspondingly, and eight optimal scenarios were then generated with different preference weights. The results based on the eight weight vectors are presented in **Figure 4**, and the objective values of each scenario are listed in **Table 3**. For the scenario of equal weights, the weight for each objective was set at 1. For the scenario preferring a specific objective, the weight for the preferred objective was 1.05 and the weights for the other objectives were set at 1.

Table 3. Optimized objective values based on various weight vectors

Optimal scenario	Objective value (normalized value)							
	Econ	Eco	H	Emp	Liv	E-H	Cpt	CC
Equal weights	39,547.03 (2.43)	63,990.73 (1.08)	14,611.36 (2.01)	19,779.92 (1.45)	25.38 (1.08)	3.69 (0.90)	53.48 (1.02)	0.10
Econ-pre	44,788.48 (2.75)	64,088.38 (1.08)	14,134.92 (1.94)	20,019.76 (1.47)	25.99 (1.11)	3.82 (0.93)	54.04 (1.04)	0.10
Eco-pre	38,877.99 (2.39)	69,760.16 (1.18)	13,878.24 (1.91)	19,010.56 (1.40)	26.74 (1.14)	3.77 (0.92)	53.27 (1.02)	0.10
H-pre	32,418.33 (1.99)	62,481.07 (1.05)	18,458.44 (2.54)	17,920.96 (1.32)	24.78 (1.05)	3.78 (0.92)	52.91 (1.01)	0.10
Emp-pre	34,611.74 (2.13)	60,582.64 (1.02)	10,403.60 (1.43)	20,190.52 (1.48)	25.06 (1.07)	4.07 (0.99)	52.49 (1.01)	0.10
Liv-pre	28,604.14 (1.76)	62,731.87 (1.06)	12,862.20 (1.77)	18,252.20 (1.34)	27.44 (1.17)	3.62 (0.88)	52.96 (1.01)	0.10
E-H-pre	37,335.43 (2.29)	63,965.00 (1.08)	14,484.36 (1.99)	19,683.32 (1.45)	25.31 (1.08)	3.77 (0.92)	53.22 (1.02)	0.10
Cpt-pre	29,097.34 (1.79)	62,117.19 (1.05)	11,846.72 (1.63)	17,428.32 (1.28)	24.61 (1.05)	3.91 (0.95)	56.19 (1.08)	0.08

^[1] A value in parenthesis is the normalized objective value that equals the absolute value divided by the value for the original scenario.

^[2] Abbreviations: Econ, Economic benefit; Eco, Ecological benefit; H, Housing capacity; Emp, Employment capacity; Liv, Livability; E-H, Employment–housing divergence; Cpt, Compatibility; CC, Conversion Cost; pre, preferred.

Table 3 shows that none of the scenarios was dominated by another scenario, but all dominated the original scenario. Specifically, all of the objective values were improved in the new scenarios relative to the original scenario. Except the E-H preferred scenario, each preferred scenario was more optimal than the other scenarios in terms of the preferred objective. For instance, the economic benefit was 44,788.48 in the economic preferred scenario, which was 2.75

times that in the original scenario and higher than that in any other scenario. For the compatibility preferred scenario, the conversion cost was 0.08, which did not reach the threshold before the 1,000th iteration.

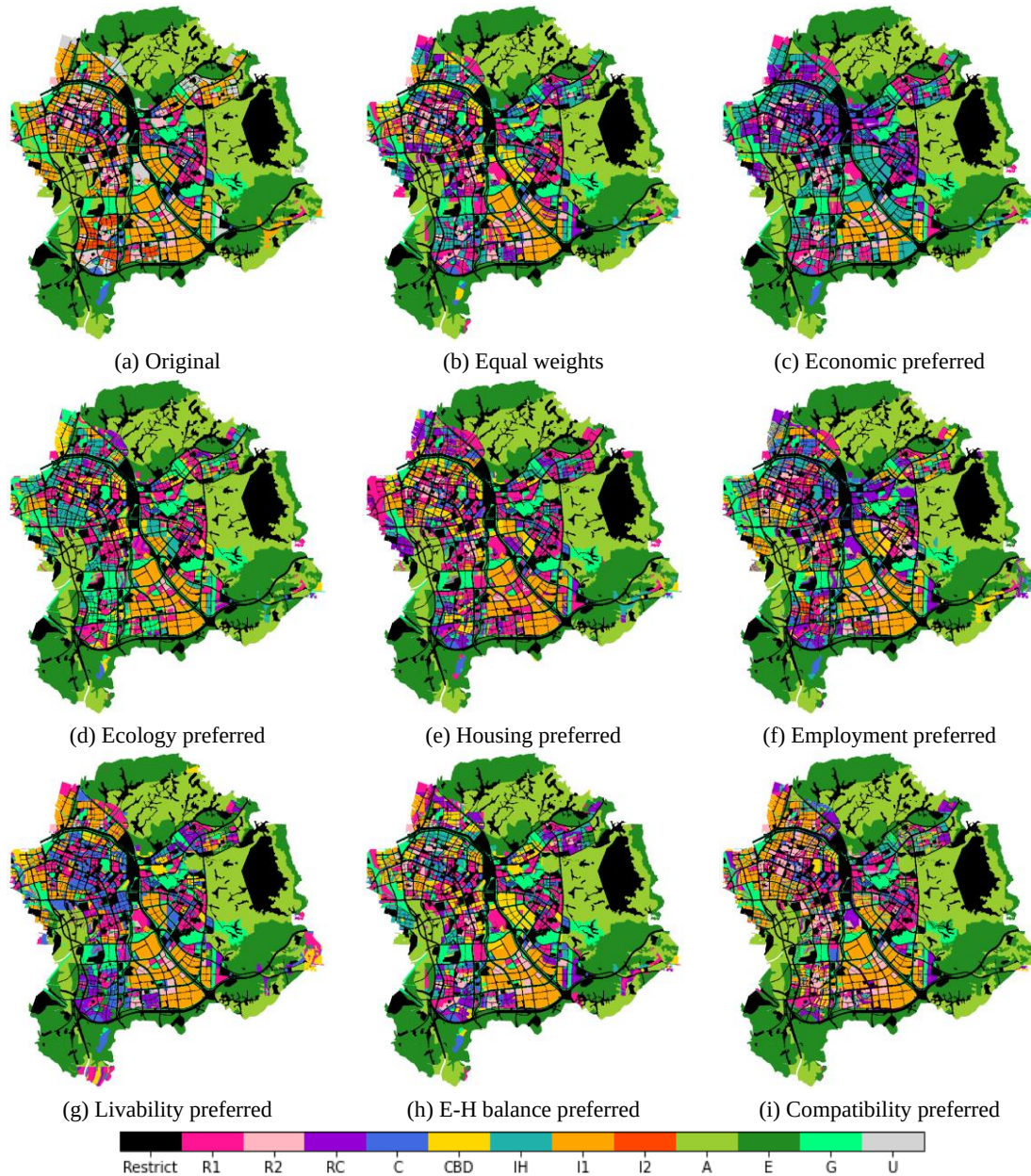


Figure 4. Optimal scenarios based on various weight vectors.

Figure 4(b) presents the equal-weights scenario, which was a satisfactory solution in general. Most of the agricultural zones and ecological zones in the outskirts of the study area were preserved. Some low-density residential zones changed to high-density zones, and some new residential regions were planned. The high-pollution industrial zones in the southwest were transformed into high-tech areas with commercial and residential zones. In the northwest and central area, low-pollution industrial zones near the downtown area became CBD zones.

The scenarios differed for different weight vectors. In **Figure 4(c)** showing the economic preferred scenario, more industrial zones are planned for high-tech industrial zones, because high-tech industrial zones are supposed to produce more economic benefits according to the hyperparameters. In **Figure 4(d)** showing the ecology preferred scenario, there are more green

lands (e.g., parks and artificial wetlands). For instance, in the southwest, the heavy pollution industrial zones are reconstructed as green lands rather than functional zones for production or residential purposes. In **Figure 4(e)** showing the housing preferred scenario, fewer low-density residential zones are preserved and more become high-density residential zones. In the west of the study area, a new high-density residential region is planned. In **Figure 4(f)** showing the employment preferred scenario, a number of pixels are converted to the CBD, high-tech industrial zones, and residential–commercial mixed zones. Additionally, several CBD zones are planned on the outskirts. In **Figure 4(g)** showing the livability preferred scenario, more commercial zones are planned near the residential zones for convenience. Additionally, some green land is allocated near residential areas. In **Figure 4(h)** showing the livability preferred scenario, more zones for production are planned near the residential areas (e.g., the CBD, high-tech industrial, and commercial zones). In this manner, the E-H divergence is reduced to improve the E-H balance. In **Figure 4(i)**, the land use map has plenty of salt-and-pepper noise, which indicates that a larger weight for compatibility is a poor choice.

6. Discussion

The major contribution of this paper has been the development of a probabilistic framework for land use optimization, in which the decision variable is a probabilistic vector and the spatial objectives are designed as differentiable objectives. The implicit spatial aspects of the problem are embedded in the objective functions such that the suitability map and convolutional operator can be applied to redefine the functions. Within the framework, the gradient-based method can thus be applied to seek solutions for large-scale spatial optimization efficiently.

Land use dynamics, one of the most concerned topics for research on sustainability, is usually explored by applying computer modeling (Ding *et al.* 2021). In previous studies, GA is the most popular due to its strong global search ability in theory. However, GA could not ensure the exact optimum and might become trapped in the local optima due to less population diversity, and inevitably faces the dilemma between convergence and stochasticity due to the nature of heuristics (Naqvi and Shad 2022). Therefore, for a large-scale problem, it is challenging to balance the population diversity and selection pressure for GA. Our experiments using simulation data also show that the computational time of the GA is $O(n)$ times that of the GDA, which concurs with the study by Nesterov and Spokoiny (2017).

Excluding the pixels outside the study area or in the restricted area, there were a total of 302,422 pixels to be optimized in the case study. In other words, the solution space was extremely high dimensional. For gradient-free algorithms, a long time is required for convergence in such a problem with a large search space. With multiple objective functions, the problem is much more complex. The GDA has been widely applied to parameter optimization for deep learning models, which shows that the GDA has the capacity to deal with optimization with numeric parameters. Thus, in practice, the GDA is a more effective method for generating a fine-resolution land use scenario within an acceptable time. The results of the simulation data experiment showed the disadvantage of the GA in solving a problem with a large solution space. In contrast, the GDA was stably effective and efficient for optimization with numerous parameters.

Like other heuristic algorithms, it is intractable to evaluate the optimality of the generated solutions by comparing them with the exact Pareto front. Even though for a small image with 8×8 pixels for the simulation data experiment, we still need 4^{64} iterations to calculate the exact optimum. However, in the experiments and case study, GDA can find solutions that significantly Pareto-dominates the original scenario. The results reveal the better performance of GDA

compared with the GA, which is meaningful in real-world applications. In practice, the time-saving framework can make it easier for users to tune the hyperparameters. With different preference weight vectors, the generated scenarios vary. Generally, a higher weight for an objective leads to more consideration of that objective by the optimizer. It takes a short time to generate an optimal solution in our framework, and planners can thus adjust the preference weight step by step to get a suitable scenario in an interactive manner. For instance, the planner can generate an equal-weights solution first and then tune the preference weights based on the equal-weights solution to get another result. The planner can keep tuning the weights according to the previous solution until a satisfactory scenario is generated.

It is noted from **Table 3** that the absolute values of various objectives range differently. In the loss function, the distances between absolute values and utopia values are normalized to the same ranges. In this study, the utopia value for each objective was estimated to be the optimized value of single-objective optimization with GDA. The strategy is intuitive but does not ensure that the found ideal value is globally optimal. Additionally, it takes some time to estimate the optimal value. In fact, all of the values in the superior (upper or lower) bound can be set as the utopia value. As an example, for an objective f to be maximized, if $a > f$ always holds true, then a can be used as the utopia value for f , although a is not the supremum. The value of the utopia point affects the weighting sensitivity, and it is ideal to set it as the infimum or supremum of objective functions. However, it is challenging to seek the infimum or supremum of a complex objective, especially in real-world applications. Hence, users can directly set a relatively high value for the utopia point or estimate the single-objective optima using GDA. With the heuristic techniques, GDA is able to find a nearly optimal solution efficiently. Even though a sub-optima is assigned as the utopia point, the algorithm is still meaningful since it can still generate a solution dominating the original situation.

Nevertheless, it is noted that the sensitivity is not the same for the preference weights of various objectives. In **Table 3**, although the preference weight of the preferred objective is set at 1.05 in various scenarios, the improvements in the preferred objective are different. That is to say, the same change in the preference weight causes different improvements for various objectives. As an example, the economic benefit in the economic preferred scenario is 1.06 times that in the equal-weights scenario, whereas the preferred objective improves by a factor of 1.29 in the housing preferred scenario. The sensitivity of preference weights is affected by the normalization strategy and varies in different cases. Hence, when tuning the preference weights, the planner should first test the sensitivity of each objective.

There are limitations to this research. Although the generated scenarios are adequate, vertically or horizontally straight boundaries and salt-and-pepper pixels can be observed. To solve this problem, a Gaussian kernel for convolution and landscape indices (e.g., compactness) could be applied in the objective function. Additionally, more characteristics of spatiality should be considered when defining the objective functions for a more practical solution scenario. Another limitation, which is also a deficiency of existing spatial optimization models, is the massive predefined hyperparameters (e.g., the weights). In empirical practice, it is difficult to define the parameters, and the parameters given by planners tend to be subjective, such that a model with fewer hyperparameters is worthy of future study.

7. Conclusion

With the expansion and development of urban areas, the land use planning of a city will cover an increasingly large area, and the grain of spatial units will become finer. Additionally, the rise of

big data brings about more sophisticated objectives. In this context, the efficiency of the framework becomes increasingly important. To deal with such challenges, we provide a new perspective for research on land use optimization by developing an efficient and effective framework with the pMOLU model and GDA. The primary purpose of the framework is to efficiently and effectively generate the optimal solution for large-scale land use optimization with massive spatial units and sophisticated objectives. We therefore proposed a pMOLU model using the continuous probability vector instead of the discrete category as the decision variable for land use representation. With the pMOLU model, the optimization objectives were designed as differentiable functions, so that the optimization problem was converted into a continuous one. The min-max utility function was adopted to combine multiple objective functions into a loss function. Adam, a popular GDA-based optimizer, was then used to minimize the loss function.

An experiment based on simulation data was conducted to compare the discrete framework with the GA and the proposed probabilistic framework with the GDA. The results show that the proposed framework was more effective and efficient, especially when the problem had a large solution space. The local gradient-based optimization method, or specifically the GDA, is thus suitable for spatial optimization. Moreover, the framework was successfully applied to the real-world case of the Guangming district of Shenzhen. The convergence progress and obtained scenarios indicated that the framework could generate a satisfactory land use planning solution in a short time. It is concluded from the results of the experiment and case study that the proposed framework can generate an optimal land use scenario in an acceptable time to support urban planning and land use management.

The development of efficient and effective land use optimization models and optimization algorithms remains a topic of concern. It is concluded from the results of the present study that gradient-based optimizers are worth applying to solving spatial optimization problems. Another direction of future work is to develop a spatiotemporal land use optimization method considering the process of land use change, instead of generating a one-shot scenario. Moreover, research on land use optimization using spatial big data is valuable in this era of big data.

Disclosure Statement

None of the authors has any conflicts of interest to declare.

Data and Codes Availability Statement

The data and codes that support the findings of this study are available with the identifier(s) at the private link: <https://figshare.com/s/1b577a983295df2fffa>

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Appendix A. Considering Diminishing Marginal Utility (DMU)

Suppose that the proportion of a specific land use type is q , and the total benefit (or utility) can be written as $U(q)$. Then, the marginal utility is $dU(q)$. In a linear model:

$$U(q) = wq,$$

where w is a weight of the corresponding land use type given by the planner, representing the average benefit utility per unit when marginal utility diminishing is not considered, so we have a constant marginal utility $dU(q) = w$.

Considering DMU, given a marginal function $u(q)$, the marginal utility could be redefined as:

$$dU(q) = wu(q)dq, 0 \leq u(q) \leq 1.$$

Then, the total utility is:

$$U(q) = \int_0^q dU(q) = w \int_0^q u(q)dq.$$

Here, we defined an equivalent benefit function $\bar{w}$, which can be considered as the average benefit utility per unit:

$$\bar{w}(w, q) = \frac{U(q)}{q} = \frac{\int_0^q u(q)dq}{q}.$$

According to the Law of DMU, the marginal function $u(q)$ should be monotonically decreasing. In addition, there should be no diminishing effect at zero points, i.e. $u(0) = 1$. In this study, the marginal function is defined as:

$$u(q) = 1 - q^2. \quad (\text{A1})$$

Then,

$$U(q) = w \int_0^q 1 - q^2 dq = w(q - \frac{1}{3}q^3).$$

Thus, the average utility, i.e. the equivalent benefit function, is:

$$\bar{w}(w, q) = w \frac{w(q - \frac{1}{3}q^3)}{q} = w(1 - \frac{1}{3}q^2).$$

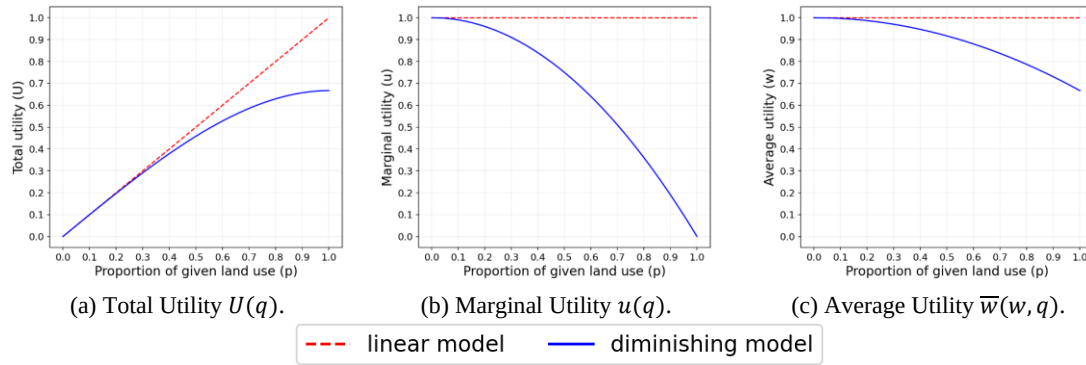


Figure A1. The function of Total Utility, Marginal Utility, and Average Utility when $w = 1$.

As **Figure A1(b)** shows, in the linear model $u(q) = 1$, which means the marginal utility is constant. Instead, in a diminishing model whose marginal function is defined by Equation (A1), the marginal utility declines as the land use percentage increases. The total utility and average utility also varied as **Figure A1 (a)** and **Figure A1 (c)** show.

Appendix B. Seeking the non-convex optimal using min-max MOUF

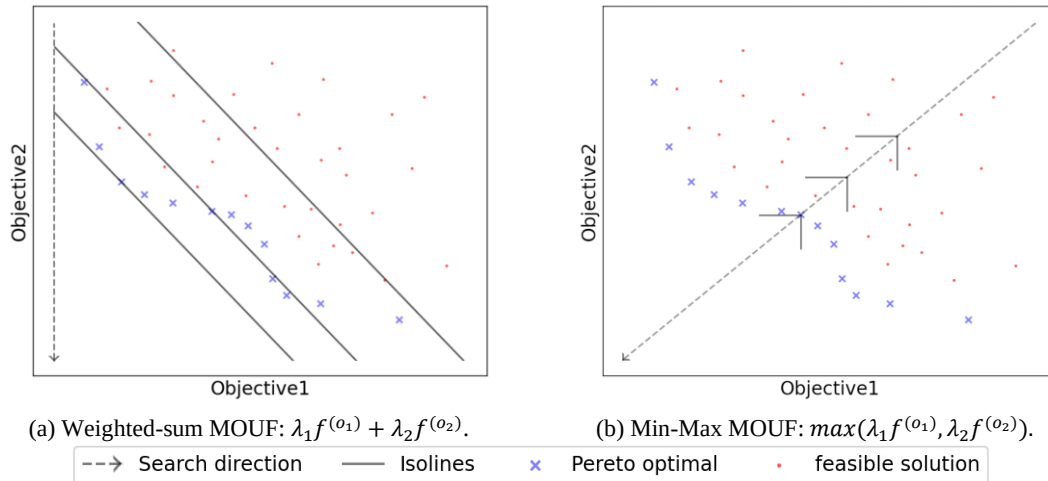


Figure B1. Seeking the optimal solution by the isolines of MOUFs.

Geometrically, **Figure B1** shows the searching direction for Pareto optimal by isolines when only two objectives are considered. For weighted sum, the isolines are a set of parallel lines: $\lambda_1 f^{(o_1)} + \lambda_2 f^{(o_2)} = C (C \in \mathbb{R}^+)$. Here, C is the value of MOUF to be minimized. As **Figure B1(a)** shows, the solution to minimizing the weighted-sum MOUF value is the only intersection point of the Pareto Front and the isoline, which is tangent to the Pareto Front. By changing the preference vector λ , the slope of lines in the isoline set is changed, and then the different optimal can be found. However, from **Figure B2(a)**, it is clear that the concave region of the Pareto Front could not be captured by the weighted-sum MOUF no matter what values the weights in the preference vector are.

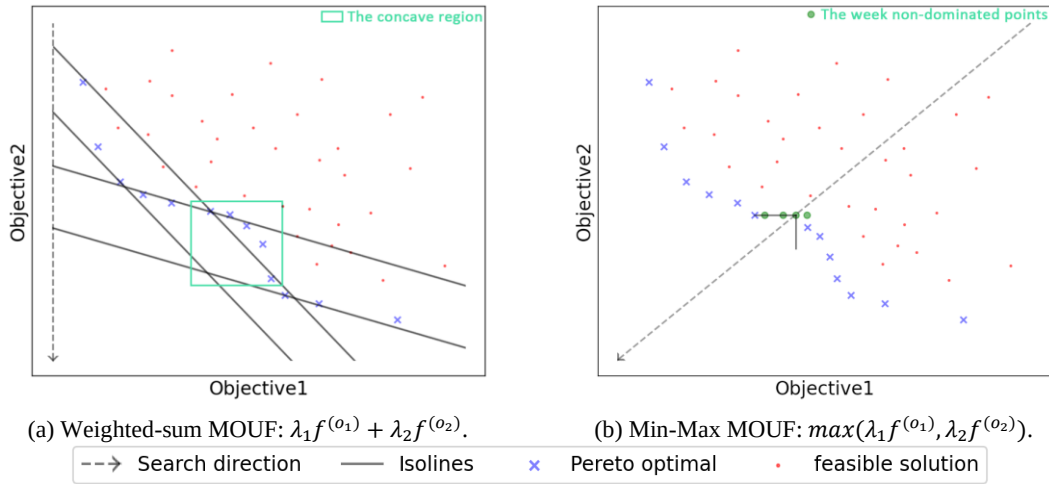


Figure B2. The disadvantage of weighted-sum MOUF.

For min-max MOUF, the searching is in the direction approaching $\mathbf{0}$ designated by the reciprocal of preference vector λ (Wierzbicki 1986). For instance, the set of isolines $\max(\lambda_1 f^{(o_1)}, \lambda_2 f^{(o_2)}) = C (C \in \mathbb{R}^+)$, is shown in **Figure B1(b)**. The vector $[-\frac{1}{\lambda_1}, -\frac{1}{\lambda_2}]$ is the searching direction, and the optimal solution for min-max MOUF is the intersection point of the feasible space boundary and the line whose direction vector is $[-\frac{1}{\lambda_1}, -\frac{1}{\lambda_2}]$ (i.e. straight line passing through the origin with slope is $\frac{\lambda_2}{\lambda_1}$). The searching direction changes as the preference vector λ changes, so the various Pareto optima point can be sought one by one. Nevertheless, the weak non-dominated solutions could be found by the min-max MOUF when the solution of minimum is not unique (Huang *et al.* 2008, Wierzbicki 1986), as **Figure B2(b)** shows. If no solution is better than solution A in all the criteria, then A is a weak non-dominated solution that is not a strict Pareto optimal solution. The weak non-dominated solution can be eliminated by comparing it with other non-dominated solutions.

Appendix C. Experiment setting

C.1 Experiment environment

The model is implemented by Python 3 with Tensorflow in a Windows operating system. The experiment is run on a machine with an Intel Core i7-7700K central processing unit (CPU) and an NVIDIA GeForce RTX 2080 Ti graphics card. Only CPU is used in the experiment using

simulation data to compare the efficiency of the two algorithms in the same environment, and GPU is used to accelerate the algorithm in the case study.

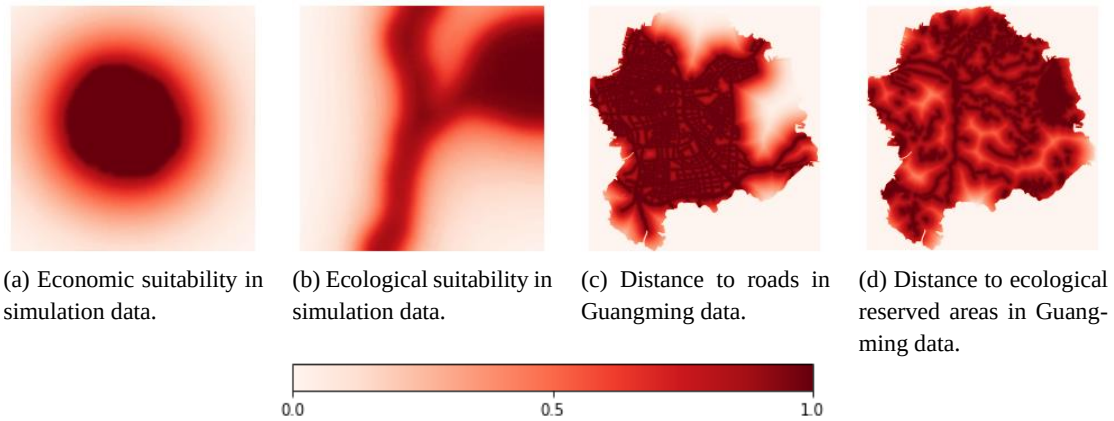


Figure C1. Simulation data and the experiment results.

C.2 Hyperparameters for the simulation data experiment

For the simulation data experiment, the coefficients for objective functions are listed in **Table C1**, and the spatial suitability images for the two objectives are shown in **Figures C1(a)** and **(b)**.

Table C1. The coefficients of land use category for objective functions.

	Urban	Rural	Industrial	Greenland
Economic coefficient	80	10	100	0
Ecological coefficient	-10	-5	-100	100

C.3 Hyperparameters for Guangming case study

Table C2 list the pre-defined weights of each land use category for various objective functions described in Section 2.2, including the economic coefficients $w^{(economic)}$, ecological coefficients $w^{(ecologic)}$, housing coefficients economic $w^{(housing)}$, employment coefficients $w^{(employment)}$ and convenient coefficients $w^{(convenient)}$. The weights are normalized such that the largest value is equal to 1.

Table C2. The coefficients of land use category for various objective functions.

Landuse	Economic coefficient	Ecological coefficient	Housing coefficient	Employment coefficient	Convenient coefficient
R1	0	0	100	0	0
R2	0	0	40	0	0
RC	35	0	30	30	10
C	100	0	0	50	100
CBD	200	0	0	90	0
IH	250	0	0	80	0
I1	50	-40	0	40	0
I2	35	-75	0	35	0
A	10	80	0	10	0
E	0	125	0	0	0
G	0	125	0	0	0

U	0	0	0	0	0
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Table C3. The compatibility matrix.

	R1	R2	RC	C	CBD	IH	I1	I2	A	E	G	U
R1	100	80	70	80	30	50	20	10	25	20	90	80
R2	80	100	55	60	40	50	50	40	30	25	90	80
RC	70	55	100	85	90	90	40	30	25	25	85	60
C	80	60	85	100	90	90	20	10	10	10	85	55
CBD	30	40	90	90	100	90	15	5	5	10	90	45
IH	50	50	90	90	90	100	15	5	25	20	90	45
I1	20	50	40	20	15	15	100	55	25	15	80	90
I2	10	40	30	10	5	5	55	100	15	5	80	90
A	25	30	25	10	5	25	25	15	100	90	30	90
E	20	25	25	10	10	20	15	5	90	100	30	90
G	90	90	85	85	90	90	80	80	30	30	100	40
U	80	80	60	55	45	45	90	90	90	90	40	100

Table C4. The conversion cost matrix.

	R1	R2	RC	C	CBD	IH	I1	I2	A	E	G	U
R1	0	30	30	30	40	40	20	20	75	90	30	30
R2	50	0	30	40	50	50	40	40	80	95	50	50
RC	50	30	0	40	50	50	40	40	80	95	50	50
C	60	40	40	0	50	50	50	50	80	95	50	50
CBD	70	50	50	50	0	30	60	60	85	95	60	60
IH	60	40	40	40	30	0	50	50	90	95	50	50
I1	25	35	35	40	50	50	0	20	95	100	30	30
I2	30	40	40	45	55	55	20	0	95	100	40	40
A	30	40	45	45	50	50	30	30	0	85	30	30
E	20	30	35	35	40	40	20	20	20	0	20	20
G	20	30	35	35	40	40	20	20	20	85	0	20
U	15	25	30	30	35	35	20	20	10	85	15	0

The $s^{(economic)}$ and $s^{(convenient)}$ mentioned in Section 2.2, is defined by the distance to the nearest road as **Figure C1(c)** shows. The $s^{(ecologic)}$ in Equation (5) is defined by the distance to the nearest water body or ecological reserved area, which is depicted in **Figure C1(d)**. Note that the values in $s^{(economic)}$, $s^{(convenient)}$ and $s^{(ecologic)}$ are normalized distances ranging from 0 to 1.

The compatibility matrix **C** mentioned in Section 2.2.7 and the transformation matrix **T** mentioned in Section 2.3.1 are defined as **Table C3** and **C4** correspondingly. The values in these two matrixes are also normalized into the $[0, 1]$ interval.

Besides, the kernel size r of the mean filter declared in Equation (8) is set to be 100, and for Adam optimizer, the learning rate is set as 0.1. In this study, the loss function is defined based on the distance to the utopia point. The utopia point is made up of the values of the optimal objective. For each objective, the best possible value is estimated by optimizing the respective objective with GDA.